

Text, Screenshots, or Both?

Representation Value and Task Routing in Web Agents

Anonymous ACL submission

Abstract

A web agent can read a page as text, look at a screenshot, or do both, and deployments usually fix one choice for every task. We measure that choice across three sites of VisualWebArena and WebArena with four vision-language backbones, using text-only variants as controls and rerunning identical conditions to see which differences survive. Three findings separate questions that are usually merged. First, no observation choice is best everywhere, and the screenshot has a real, reproducible value that is predictable in advance on one site: a zero-cost rule over the task text picks out classifieds tasks on which adding the screenshot alone raises success by 25–30 points for the two strongest backbones (11 for a 4B model), and the gap reappears on a full rerun. Second, which representation suits which task is a reproducible property of the task, beyond an additive model of task difficulty, yet a single run measures it with reliability of only 0.31–0.50. Third, two lightweight routers that choose a representation per task from pre-run features, with their operating point fixed on training data, show no reliable gain over the fixed choices and their random mixtures. We describe the rerun-controlled protocol that keeps these three quantities apart.

1 Introduction

An agent that operates a website has to perceive it, and the two standard ways are complementary in principle: the page’s text (an accessibility tree) is precise and cheap, a screenshot shows what the text omits. A third option fuses them, overlaying numbered marks on the screenshot and listing the marked elements as text (Yang et al., 2023; Koh et al., 2024). Practitioners typically pick one of the three for a whole deployment. The question this paper asks is the one that choice raises: *is the choice worth making per task, and can it be made?*

Answering it requires keeping apart three quantities that a single leaderboard number merges:

1. **Representation value:** does the screenshot change success, holding everything else fixed?
2. **Preference reliability:** when one representation beats another on a task, is that a stable property of the task, or a property of one run?
3. **Routing gain:** can a policy that chooses per task, using only what is known before it runs, beat the fixed choices?

A hindsight ceiling picks, for each task, whichever representation happened to succeed. However large, it answers none of them, because it rewards run-to-run luck as much as structure. We therefore rerun identical conditions and read every comparison against what a rerun alone changes.

Findings. (i) No choice wins everywhere: the fused mode leads on classifieds, text leads on WebArena reddit, and the order reverses across backbones (§3). (ii) The screenshot has a specific value that can be isolated and predicted: with the text payload and prompt held fixed, adding the screenshot raises success on the classifieds tasks a zero-cost intent rule flags by 25–30 points for the two strongest backbones (11 for a 4B model), against 5–10 on the other tasks; the effect reproduces on a full rerun and is larger for the stronger backbones we tested. On reddit the same rule finds nothing (§4). (iii) Per-task preferences between representations are reproducible beyond an additive difficulty model: a no-interaction null conditioned on every task’s and every mode’s success count is rejected in every fully rerun cell. Yet a single run measures them with reliability 0.31–0.50, and a decision study projects that labels averaged over two to three runs would pass 0.5 (§5). (iv) Two lightweight routers that fix their operating point on training data show no reliable gain over the fixed representations and their random mixtures; the one hindsight opportunity that passes its null comes from a router that only decides which tasks deserve the expensive

mode, holds under billed cost only, and is not captured without looking at test outcomes (§6).

Contributions. A rerun-controlled measurement of text, screenshot and fused observation across three sites and four backbones, with text-only ablations that isolate the screenshot; a matched estimate of the screenshot’s value and an ex-ante rule that predicts it on one site; a test of whether per-task representation preferences are reproducible, with a decision study of how many runs a reliable label needs; and a deployable-policy comparison against fixed choices *and their random mixtures*, which is the comparison a routing claim has to win; we take this baseline from model-routing benchmarks and apply it to observation choice (§7).

2 Setup

Sites and tasks. We use three sites of VisualWebArena (Koh et al., 2024): classifieds (224 scored tasks), reddit (203) and shopping (432). We add the reddit site of WebArena (Zhou et al., 2024) (104 tasks: the 106 minus two the benchmark marks as not applicable). VWA-reddit and WA-reddit are different task sets on the same application, so they are not two independent benchmarks. Scored sets are the collected tasks minus protocol exclusions written as rules over task configuration (Appendix A).

Backbones. B0 = Qwen3-VL-235B (API-served), B1 = Qwen3-VL-4B and B2 = Gemma-3-4B (both served locally), and B5 = GPT-5.6 (API, classifieds only). A *cell* is one site $\times$ backbone pair; the eight cells with every observation mode are VWA-classifieds and VWA-reddit $\times$ {B0, B1, B2} and WA-reddit $\times$ {B0, B1}. B5-classifieds and the two shopping cells extend them (§4, §6) and are never pooled with the eight.

One plain agent. One model call per step emits a thought, an action and its arguments; the prompt holds the instruction, the mode’s system prompt, the last eight steps and the current observation. There is no planner, memory, reflection or retry policy, so a difference between conditions is a difference of observation, not of scaffold. Decoding is greedy, the viewport is 1280×720 , the text observation is never truncated, and the nominal budget is 30 steps (one backbone’s parser let some extra actions through; Appendix E). Appendix B gives the full contract.

Observation modes. The deployment question has three answers: **DOM** (the page’s accessibility tree as text), **Vision** (the raw screenshot, with an empty text payload) and **SoM** (a set-of-marks screenshot plus the list of marked elements as text; Yang et al., 2023). Three further modes carry no screenshot and serve as controls; the one used throughout is **P-SoM**, which is SoM with the screenshot removed. Marks and prompt are the same, so SoM–P-SoM isolates the image (Appendix C lists all six). “Text” modes withhold the *page* screenshot only: a reference image that is part of the task (“find this item”) is given in every mode.

Reruns. We reran identical conditions, 24 registered same-condition pairs in all. In three cells (VWA-classifieds B0, VWA-reddit B0 and WA-reddit B1), every mode was run twice. A rerun uses the same tasks and code on freshly reset sites, so what it changes is what one run of the agent does not determine. Sites are reset before every run; within a run only classifieds resets between tasks, so reddit and shopping carry state from earlier episodes (Appendix E).

Scoring and cost. The evaluators are binary. Cost is the per-episode cost in US dollars: the API charge for B0 and B5, and for B1 and B2 a token-priced proxy for local serving (hardware amortisation plus electricity per generated token; per-cell values in Table 3). The two kinds are ordered within a cell, never pooled across; §6 repeats the routing analysis under GPU-time and wall-clock cost.

3 No observation choice is best everywhere

Figure 1 shows the three deployment choices side by side. On classifieds the fused mode is the best of the three for B0, B1 and GPT-5.6 (B2 solves too few tasks to order the modes), and the margin over text is not small: SoM–DOM is +9.82 pp [+3.57, +16.07] for B0 and +8.04 pp [+3.57, +12.95] for B1 (paired task bootstrap). On WebArena reddit the order flips and text leads (SoM–DOM –4.81 pp for B0 and –2.88 pp for B1, both intervals crossing zero), and on VWA-reddit the weakest backbone does worse with the fused mode (–2.96 pp [–5.91, –0.49]), but three of its eight DOM successes were credited although the episode never visited the target forum, whose subscription an earlier episode had made; without them the gap is

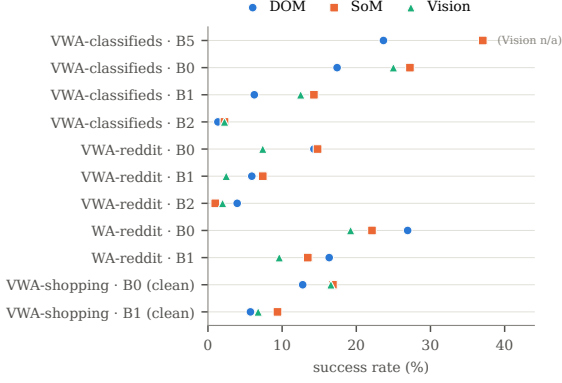


Figure 1: Success rate of the three deployment choices per cell. Shopping cells drop the tasks touched by a known harness defect in any mode (§2, Appendix E); B5’s Vision run is excluded for a coordinate-contract defect. Source: routing_three_arm, routing_extension_cells.

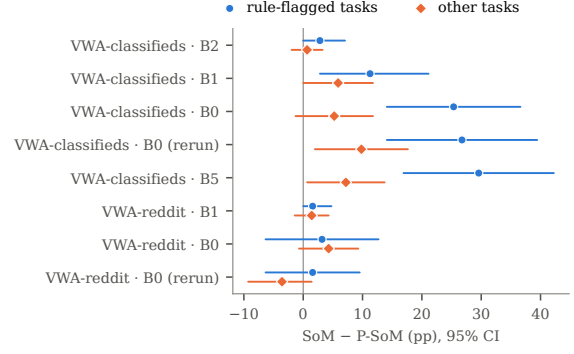


Figure 2: Value of the screenshot alone (SoM – P-SoM: same marks, same prompt) on the tasks a zero-cost intent rule flags, against the other tasks. Classifieds is out-of-sample for the rule. “Rerun” is a full repeat of both modes. VWA-reddit B2 is omitted: neither SoM nor P-SoM solves a flagged task. Source: visual_intent_routing.

–1.48 pp [–3.45, +0.49] (Appendix E). On shopping the fused mode is best or tied with Vision, on the filtered and on the full task sets (Figure 1, Table 3).

Two cautions keep this descriptive. Cells on the same site share their tasks, so the eight are not eight independent replications. And the screenshot-free modes are four, not one: a maximum over four text variants against a single screenshot mode would be a selection effect, so the comparison here takes one representative per deployment class, as practitioners do.

What this section establishes is only that the choice matters and depends on the workload. Whether it can be made per task is the subject of the rest of the paper.

4 The screenshot has a value that can be isolated and predicted

A matched contrast. Comparing Vision with DOM changes two things at once: the image and the text the agent receives. SoM and P-SoM differ only in the annotated screenshot, so their difference is the value of the image alone. Chi et al. (2024) already compared screenshot-plus-text with text only on a subset of VisualWebArena tasks for one backbone (50.0% against 46.3%, without set-of-marks); we hold the set-of-marks text and prompt fixed, use four backbones and full task sets, rerun the contrast, and ask *where* the image pays off.

An ex-ante rule. A regular expression over the task intent flags tasks that ask about an image, a

photo, a colour or a count of things shown, excluding tasks that already carry a reference image (Appendix D). It reads only the task configuration: no model call and no episode. It was written as a reddit failure-diagnosis pattern a week before it was first used to partition tasks, and its classifieds hits were never inspected, so classifieds is out-of-sample for it.

Result. On the 71 classifieds tasks it flags, the screenshot raises success by +25.35 pp [+14.08, +36.62] for B0, against +5.23 pp on the other 153 tasks (Figure 2). The effect reproduces on a full rerun of both modes (+26.76 pp [+14.08, +39.44], from 23 against 4 successes), and it is larger for the stronger backbones we tested: +2.82 pp for B2, +11.27 pp for B1, +25.35 pp for B0 and +29.58 pp [+16.90, +42.25] for GPT-5.6 (whose parser let some actions through beyond the step budget, in both modes alike; Appendix E). The backbones differ in more than capability, so this is an association across the four we ran, not a scaling law. On VWA-reddit the same rule flags 63 tasks and separates nothing: +3.17 pp on flagged tasks against +4.29 pp elsewhere for B0, +1.59 pp against –3.57 pp on its rerun. On shopping the flagged tasks carry at most five successes per mode, too few to read, and on WebArena the rule flags only 5 of 104 tasks.

Reading. Where the screenshot pays off on classifieds is predictable before anything runs, and it pays off more for stronger models; on reddit it is not. The screenshot’s value is therefore real and partly foreseeable, but it belongs to a kind of task

on one site. It is also not a router. On these tasks the fused mode already contains the screenshot, so switching between DOM and Vision by this rule cannot beat simply using SoM; what remains open is whether *individual* tasks have stable preferences beyond such task types, which is the question of §5.

5 Per-task preferences are real, but one run cannot measure them

§4 found structure at the level of task *types*. Routing per task needs more: that an individual task prefers one representation over another, beyond being easy or hard, and that this preference is stable enough to learn from.

A null without task-by-mode structure. In the three cells where every mode was run twice, let $Y_{imr} \in \{0, 1\}$ be the outcome of task i under mode m in run r . Under

$$\text{logit } \Pr(Y_{imr} = 1) = a_i + b_{mr},$$

every task has its own difficulty and every (mode, run) column its own easiness, but there is no persistent task-by-mode interaction. This is the Rasch structure that item-response routers use to model difficulty (Song et al., 2025); here it is the null, and the question is what lies beyond it. The row and column sums are sufficient for this model, so conditional on them every 0/1 matrix with those margins is equally likely; we approximate that conditional null by curveball Markov-chain sampling (Strona et al., 2014), fitting nothing. The statistic is the reproducible part of the interaction, $\hat{\sigma}_{\text{int}}^2 = (\text{MS}_{\text{int}} - \text{MS}_{\text{err}})/2$ from the two-way task $\times$ mode ANOVA with two runs: the run-to-run covariance of each task’s mode profile.

Result. Over the three deployment modes the interaction exceeds the null in all three cells (classifieds B0 $\hat{\sigma}_{\text{int}}^2 = 0.033$ against a null 95th percentile of 0.015; reddit B0 0.018 against 0.009; WebArena-reddit B1 0.030 against 0.014; $p \leq 0.002$ each, Holm across cells). Over all six modes the same holds (Appendix F). Preferences between representations are therefore a reproducible property of tasks that an additive difficulty model does not capture. The test does not say what the extra structure is: a task-specific preference and a mode whose success responds more steeply to difficulty than another’s both reject this null, and only the first is a representation preference in the narrow sense.

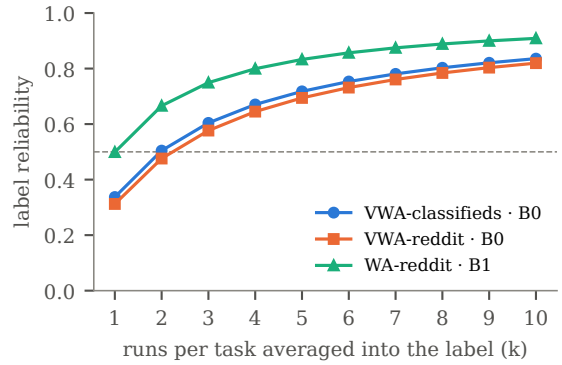


Figure 3: Decision study for the three deployment modes: reliability of a task-by-mode label averaged over k runs, $\sigma_{\text{int}}^2 / (\sigma_{\text{int}}^2 + \sigma_{\text{err}}^2/k)$. A single run gives 0.31–0.50. Source: routing_three_arm.

But a single run measures them poorly. The same decomposition gives, in the sense of a generalisability decision study (Brennan, 2001), the reliability of a task’s mode profile averaged over k runs (Figure 3). From one run it is 0.34, 0.31 and 0.50 in the three cells; the projection crosses 0.5 at two to three runs (bootstrap ranges 2–4, 2–7 and 1–3). Over six modes it is lower still, 0.14–0.46, with up to six runs projected. This is the reliability of the interaction profile, not of a router’s chosen arm, so it bounds how noisy single-run training labels are rather than predicting when a router starts to win. The preferences also persist where they can be observed directly: over all six modes, taking for each task the cheapest mode that succeeded on one run and scoring that choice on the rerun beats the fixed-mode frontier by 2.4–10.5 points on reddit B0 and WebArena-reddit B1. That choice uses the task’s own label, which no deployment has, so it measures persistence, not an attainable policy.

The routing target thus exists, and a training set built from single runs labels it with a reliability between a third and a half.

6 Deployable routers do not reliably convert the signal

The comparison a router must win. A router that sends some tasks to an expensive mode and others to a cheap one trades success for cost. Beating every fixed mode on both axes is too strict, but beating the fixed modes one at a time is too lenient: sending a random share of tasks to each of two modes is also a deployable policy. We therefore compare against the *frontier of the fixed modes and all their random mixtures*: the upper concave

envelope of the fixed points in the (cost, success) plane, which is the non-decreasing convex hull that model-routing benchmarks use as a baseline (Hu et al., 2024; Jitkrittum et al., 2025). We report how far above it a router reaches (Figure 4a; definitions in Appendix I).

Routers. Two constructions use 18 features available before a task runs: 14 intent-keyword indicators, the intent length, and three statistics of the first page. *Per-arm heads* fit one logistic success model per mode and send a task to the cheapest mode whose predicted success clears a threshold. *Triage* predicts whether any mode can solve the task, sends predicted-hopeless tasks to the cheapest mode and the rest to the best one. All predictions are out-of-fold (task-held-out five-fold).

Room above the frontier exists in hindsight. Sweeping each router’s threshold traces a curve. Over the three deployment modes, the triage curve pooled over the eight cells rises 1.69 points above the frontier at low budget (Figure 4b), which a label-shuffle null that refits the whole curve does not produce ($p = 0.001$; Table 1). The per-arm heads reach 0.86 points ($p = 0.072$). Triage does not choose a representation for a task; it decides which tasks are not worth the expensive mode. That one construction passes its null and the other does not is not a test that representation fit carries nothing, since a difference in significance is not a significant difference. In these data, however, the only opportunity that clears a null is the one a difficulty-only decision exposes, and it does so under billed cost only (see Robustness).

A deployable router does not capture it. A curve’s best point is chosen after seeing test outcomes. When the operating point is instead chosen on training folds only, the pooled gain is +0.23 points for per-arm heads and +0.59 for triage, and neither interval excludes zero (Table 1). Over all six modes both deployable routers sit slightly below the frontier (−0.38 and −0.39). The pattern holds for the strongest backbone: for GPT-5.6 on classifieds, where the screenshot’s value was largest (§4), the deployable routers lose 1.6–2.5 points; on both shopping cells, filtered or on all tasks, they stay within 1.3 points of it.

Robustness. Holding tasks of the same template out together lowers the six-mode per-arm router’s mean deployable gain from −0.38 to −1.60 points, and fitting on one run and scoring on its rerun

gives point estimates of both signs whose intervals all include zero (Appendix G). Repricing cost changes the hindsight result but not the deployable one (Table 2). Under wall-clock cost the pooled triage curve no longer clears its null (+0.68 points, $p = 0.42$), nor does it under GPU time on the five locally served cells (+1.24, $p = 0.39$); the per-arm heads stay just short of significance under every basis ($p = 0.072$ – 0.077). Over all cells of each basis the deployable gains stay between −0.5 and +1.5 points. Dropping the two cells of the weakest backbone, which solves too few tasks for any router to gain, leaves the billed picture unchanged (triage curve +1.85, $p = 0.001$; deployable +0.50 and +0.39). One interval in Table 1 (per-arm heads, three modes pooled) has an upper end below its point estimate, a sign that the task bootstrap of a policy whose threshold is re-selected in every draw is biased; we therefore read all deployable intervals as magnitudes and make no significance claim from them, in either direction.

7 A protocol for multi-representation claims

The three quantities of §1 came apart because each was measured against its own control. We suggest the same four steps for any claim that choosing among several agent configurations per task would pay.

1. **Report reruns per arm and per cell**, not one flip rate or one resolution threshold for a whole study: in our data a rerun changes between 0 and 14% of outcomes depending on arm and cell (Appendix F).
2. **Isolate the component of interest with a matched ablation** (here SoM–P-SoM, which removes only the screenshot) before attributing a difference to it.
3. **Test for per-task structure against a null that conditions on difficulty**, and use a decision study to say how many runs per task make the label reliable. A single-run oracle is not a ceiling a router can reach.
4. **Score a deployable policy, operating point chosen on training data, against the fixed arms and their random mixtures**, never only against the best single arm or a hindsight curve.

Applied here, the protocol says the screenshot

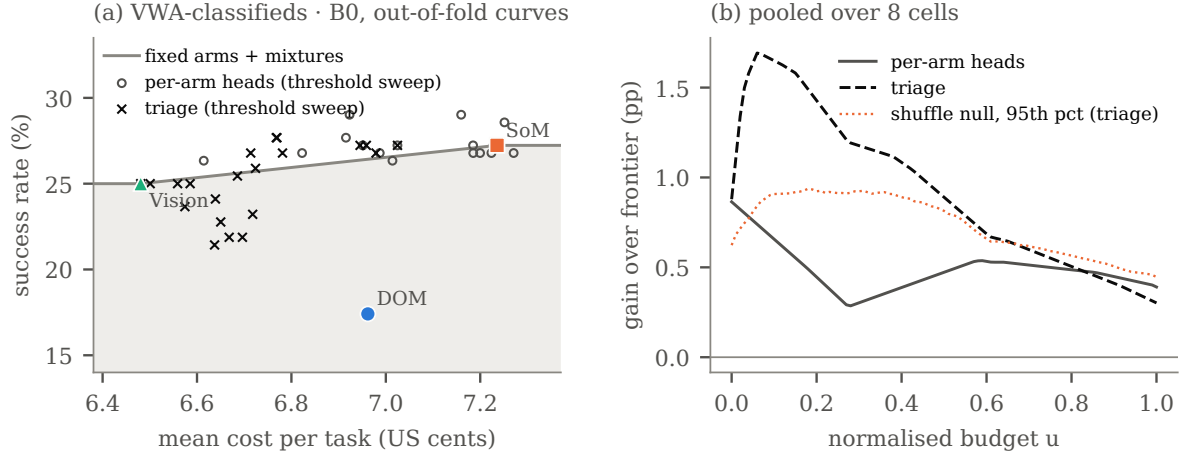


Figure 4: (a) One cell’s cost–success plane: the three deployment arms, the frontier of the arms and their random mixtures (shaded: attainable without routing), and the out-of-fold points of the two routers as their threshold is swept. (b) Gain of the frontier when each router’s curve is added, averaged over the eight cells on a budget normalised so that the cheapest arm is $u = 0$ and the dearest $u = 1$, against the 95th percentile of a label-shuffle null that refits the whole curve. Both panels choose the best point after seeing test outcomes; Table 1 gives the deployable version. Source: routing_three_arm.

is worth paying for on a predictable class of classifieds tasks, that per-task representation preferences are reproducible but single-run labels measure them with reliability of a third to a half, and that with such labels and pre-flight features the routers we tested do not reliably beat choosing one representation, or a fixed mixture, for the whole workload.

8 Related work

How web agents observe the page. Agents have read HTML or accessibility trees (Deng et al., 2023; Yang et al., 2025), screenshots grounded to elements (Zheng et al., 2024; He et al., 2024; Gou et al., 2025), or set-of-marks overlays that fuse the two (Yang et al., 2023; Koh et al., 2024). A parallel line shrinks or reshapes the text observation (Kerboua et al., 2025; Schiepanksi and Piël, 2025; Enomoto et al., 2026), and some systems combine several grounding routes (Li et al., 2026a). OSWorld compares screenshot, accessibility-tree, combined and set-of-marks inputs across models and finds the best input model-dependent (Xie et al., 2024), and Chi et al. (2024) compare screenshot-plus-text with text only on a VisualWebArena subset. Closer to adaptivity, Li and Shi (2026) gate visual observation components during execution of computer-use tasks, and Liu et al. (2026) route each action of a computer-use agent among vision-language models; neither chooses between text and screenshot per task before it runs. These works

compare observation designs at the level of a benchmark score or adapt them online. Our contribution is not that the best observation varies, which OSWorld already shows, but a matched removal of the screenshot, repeated task-level outcomes, a test of task structure beyond difficulty, and a deployable per-task choice judged against mixtures of the fixed choices.

Routing and cascades. Choosing among models per query to trade quality for cost is well studied for language models (Chen et al., 2024; Ong et al., 2025; Ding et al., 2024; Gupta et al., 2024; Peale et al., 2026); see Moslem and Kelleher (2026) for a survey. For web agents, Li et al. (2025) learn query-specific routing for cost-sensitive agents, and Hajimiri et al. (2026) judge agent components at a matched budget. Early exit (Lu et al., 2025) stops agents that are unlikely to succeed. That is the difficulty lever, which in our data is the only part of routing that leaves room above the fixed frontier. Routing benchmarks compare routers against random mixtures of fixed models (Hu et al., 2024; Jitkritum et al., 2025), and under unified conditions several learned routers fail to beat a simple baseline (Li et al., 2026b); Lee (2026) finds that assigning one model per task type in advance recovers most of the oracle routing gain on question answering, and that reruns rescore 5% of model-question pairs. Item-response routers model query difficulty and model ability (Song et al., 2025). We

cells	policy	curve max (p)	deployable [5%, 95%]
3 arms, 8 cells pooled	per-arm heads	+0.86 (0.072)	+0.23 [−2.20, +0.22]
	triage	+1.69 (0.001)	+0.59 [−1.53, +1.23]
6 arms, 8 cells pooled	per-arm heads	+0.67 (0.252)	−0.38 [−3.20, −0.14]
	triage	+1.24 (0.043)	−0.39 [−2.42, +0.81]
GPT-5.6, classifieds (5 arms)	per-arm heads	+0.45 (0.768)	−2.45 [−8.61, +4.24]
	triage	+3.86 (0.127)	−1.63 [−6.70, +5.74]
Shopping B0, clean (3 arms)	per-arm heads	+1.14 (0.062)	−0.18 [−2.45, +2.02]
	triage	+0.20 (0.496)	−0.87 [−2.63, +1.37]
Shopping B0, all tasks (3 arms)	per-arm heads	+1.09 (0.049)	+1.30 [−2.31, +1.85]
	triage	+0.18 (0.541)	−0.05 [−2.14, +1.45]
Shopping B1, clean (6 arms)	per-arm heads	+1.11 (0.143)	+0.56 [−4.69, +0.88]
	triage	+1.63 (0.158)	+1.19 [−4.10, +1.57]
Shopping B1, all tasks (6 arms)	per-arm heads	+0.48 (0.169)	−0.43 [−2.78, +0.73]
	triage	+0.69 (0.289)	+0.40 [−2.02, +2.02]

Table 1: Routing gain over the fixed arms and their random mixtures, in SR points. *Curve max*: the best point of an out-of-fold curve, chosen after seeing test outcomes, with its label-shuffle p (pooled rows: max over a normalised budget). *Deployable*: the operating point chosen on training folds only, task bootstrap. Sources: routing_three_arm, representation_routing_frontier, routing_gain_upper_bounds, routing_extension_cells.

route over observation representations rather than models, take the mixture baseline from this literature, and add what it rarely has: reruns of identical agent conditions and a test of per-task structure beyond difficulty.

Reliability of agent evaluation. Repeated trials of the same agent disagree, which τ -bench measures with pass^k (Yao et al., 2025); inference itself is not deterministic even at fixed settings (He, 2025; Yuan et al., 2025), prompts change outcomes in ways unrelated to content (Sclar et al., 2024), and agent benchmarks carry evaluator and task defects (Hattami et al., 2025; Bhat et al., 2026). Our reruns quantify the first for web agents at the task level and show what it does to per-task routing labels; the harness defects we found are listed in Appendix E.

Limitations

Coverage of reruns. Only three cells have every mode run twice, two with the same backbone; the preference test and decision study (§5) rest on them. Five of the eleven cells have no rerun of any mode.

Breadth. VWA-reddit and WA-reddit are task sets on the same application, so the study covers three applications, not four benchmarks. GPT-5.6 was run on classifieds only, and its Vision run is excluded for a coordinate-contract defect; a parser defect also let it take actions beyond the 30-step budget in every mode (Appendix E). The ex-ante rule of §4 works on classifieds only.

Shopping filtering. The clean shopping cells drop every task on which the search-box defect fired in any mode. Whether it fired depends on what the agent typed, so the filter selects on behaviour under the conditions being compared: the clean cells estimate the comparison on an observed defect-free subset, not on the benchmark, and B0 and B1 keep different subsets (368 and 192 tasks). Results on all tasks are reported alongside (Tables 1 and 3); no conclusion rests on the filtered cells alone.

What the routing result does not show. The routers use 18 pre-flight features and logistic models; richer features, language-model routers or multiple runs per training task may do better. The result is that, with single-run labels and these features, the routers do not beat the fixed choices and their mixtures. It does not show that per-task routing is impossible. Step-level switching is out of scope.

Inherited state. Reddit and shopping are reset before each run but not between tasks, so an episode can be credited for state an earlier episode created. A mechanical audit removes reddit successes credited without visiting the target forum and changes one comparison in the paper (§3); it cannot see an episode that visits a forum an earlier episode already subscribed to and leaves without acting, so reddit outcomes may still carry inherited credit, and reruns may reproduce it.

Exploratory analyses. The preference test, frontier comparisons and bounds were designed after

the runs were collected. Each family states its multiplicity correction; the bootstrap intervals for policies with a selected threshold are read as magnitudes, not tests.

Cost. API invoices and electricity estimates are different quantities and are never pooled; local cost depends on the pricing basis, which is why \$6 repeats the analysis under GPU time and wall-clock time.

Ethics statement

All episodes ran against self-hosted copies of the VisualWebArena and WebArena sites; no live website, real account or person was contacted, and no human subjects were involved. Sites were reset before each run. The API-served backbones were used under their providers’ terms. Agents that act on websites can cause harm when deployed against live services; this study measures observation choices in sandboxes and releases no agent that acts outside them.

Availability. The anonymous supplementary material contains the agent and run configurations, per-episode outcomes and costs for every run used, the analysis scripts, and every analysis product cited in the text, with a guide that maps each number in the paper to the product field it comes from; figures and tables are regenerated from the products by one script. Full trajectories (screenshots and page text) are too large for the supplement and will be released with the camera-ready version.

References

Vishvesh Bhat, Jay Vaghasiya, Muhammad Ahmed Mohsin, and Asad Aali. 2026. [Benchmarking the Benchmarks: A Validity Audit of Tool-Calling Evaluation](#). *Preprint*, arXiv:2607.02577.

Robert L. Brennan. 2001. *Generalizability Theory*. Springer, New York.

Lingjiao Chen, Matei Zaharia, and James Zou. 2024. [FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance](#). *Transactions on Machine Learning Research*.

Wayne Chi, Ameet Talwalkar, and Chris Donahue. 2024. [The impact of element ordering on LM agent performance](#). *Preprint*, arXiv:2409.12089.

Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Sam Stevens, Boshi Wang, Huan Sun, and Yu Su. 2023. [Mind2Web: Towards a Generalist Agent for the Web](#). In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*.

Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Rühle, Laks V. S. Lakshmanan, and Ahmed Hassan Awadallah. 2024. [Hybrid LLM: Cost-Efficient and Quality-Aware Query Routing](#). In *International Conference on Learning Representations*.

Masafumi Enomoto, Ryoma Obara, Haochen Zhang, and Masafumi Oyamada. 2026. [Revisiting Observation Reduction for Web Agents: Comprehensive Evaluation with a Lightweight Framework](#). *Preprint*, arXiv:2605.29397.

Boyu Gou, Demi Ruohan Wang, Boyuan Zheng, Yanan Xie, Cheng Chang, Yiheng Shu, Huan Sun, and Yu Su. 2025. [Navigating the Digital World as Humans Do: Universal Visual Grounding for GUI Agents](#). In *International Conference on Learning Representations*.

Neha Gupta, Harikrishna Narasimhan, Wittawat Jitkrittum, Ankit Singh Rawat, Aditya Krishna Menon, and Sanjiv Kumar. 2024. [Language Model Cascades: Token-Level Uncertainty and Beyond](#). In *International Conference on Learning Representations*.

Sina Hajimiri, Masih Aminbeidokhti, Jose Dolz, Ismail Ben Ayed, Issam H. Laradji, Spandana Gella, and Nicolas Gontier. 2026. [Are Online Skill and Memory Modules Always Worth Their Tokens? A Budget-Constrained Study of Web Agents](#). *Preprint*, arXiv:2606.15017.

Amine El Hattami, Megh Thakkar, Nicolas Chapados, and Christopher Pal. 2025. [WebArena Verified: Reliable Evaluation for Web Agents](#). In *Workshop on Scaling Environments for Agents*. NeurIPS 2025 workshop; also see <https://github.com/ServiceNow/webarena-verified>.

Hongliang He, Wenlin Yao, Kaixin Ma, Wenhao Yu, Yong Dai, Hongming Zhang, Zhenzhong Lan, and Dong Yu. 2024. [WebVoyager: Building an End-to-End Web Agent with Large Multimodal Models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

Horace He. 2025. [Defeating Nondeterminism in LLM Inference](#). Thinking Machines Lab blog.

Qitian Hu, Jacob Bieker, Xiuyu Li, Nan Jiang, Benjamin Keigwin, Gaurav Ranganath, Kurt Keutzer, and Shriyash Upadhyay. 2024. [RouterBench: A benchmark for multi-LLM routing system](#). In *Agentic Markets Workshop at ICML 2024*.

Wittawat Jitkrittum, Harikrishna Narasimhan, Ankit Singh Rawat, Jeevesh Juneja, Congchao Wang, Zifeng Wang, Alec Go, Chen-Yu Lee, Pradeep Shenoy, Rina Panigrahy, Aditya Krishna Menon, and Sanjiv Kumar. 2025. [Universal model routing for efficient LLM inference](#). *Preprint*, arXiv:2502.08773.

644	Imene Kerboua, Sahar Omid Shayegan, Xing Han	Charlotte Peale, Siddhartha Devic, Parikshit Gopalan,	697
645	Lù, Léo Boisvert, Megh Thakkar, Massimo Cac-	Udi Wieder, and Aravind Gollakota. 2026. Flexible	698
646	cia, Jérémy Espinas, Alexandre Aussem, Véronique	Routing via Uncertainty Decomposition . <i>Preprint</i> ,	699
647	Eglin, and Alexandre Lacoste. 2025. FocusAgent:	arXiv:2605.07805. ICML 2026 Workshop on Statisti-	700
648	Simple yet effective ways of trimming the large con-	cal Frameworks for Uncertainty in Agentic Systems.	701
649	text of web agents . <i>Preprint</i> , arXiv:2510.03204.		
650	Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram	Thassilo M. Schiepanski and Nicholas Piël. 2025.	702
651	Duvvur, Ming Lim, Po-Yu Huang, Graham Neu-	Beyond Pixels: Exploring DOM Downsam-	703
652	big, Shuyan Zhou, Russ Salakhutdinov, and Daniel	pling for LLM-Based Web Agents . <i>Preprint</i> ,	704
653	Fried. 2024. VisualWebArena: Evaluating Multi-	arXiv:2508.04412.	705
654	modal Agents on Realistic Visual Web Tasks . In	Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane	706
655	<i>Proceedings of the 62nd Annual Meeting of the Asso-</i>	Suhr. 2024. Quantifying Language Models' Sensitiv-	707
656	<i>ciation for Computational Linguistics</i> .	ity to Spurious Features in Prompt Design or: How	708
657	Janghoon Lee. 2026. Most of the LLM routing gap is	I learned to start worrying about prompt formatting .	709
658	task type . <i>Preprint</i> , arXiv:2608.23023.	In <i>International Conference on Learning Representa-</i>	710
659	Aiden Yiliu Li, Xinyue Hao, Shilong Liu, and Mengdi	<i>tions</i> .	711
660	Wang. 2026a. Avenir-Web: Human-Experience-	Wei Song, Zhenya Huang, Cheng Cheng, Weibo Gao,	712
661	Imitating Multimodal Web Agents with Mixture of	Bihan Xu, GuanHao Zhao, Fei Wang, and Runze	713
662	Grounding Experts . <i>Preprint</i> , arXiv:2602.02468.	Wu. 2025. IRT-Router: Effective and interpretable	714
663	Bojie Li and Noah Shi. 2026. Agent-computer ob-	multi-LLM routing via item response theory . In <i>Pro-</i>	715
664	servation interfaces enable dynamic computer use .	<i>ceedings of the 63rd Annual Meeting of the Associa-</i>	716
665	<i>Preprint</i> , arXiv:2606.29472.	<i>tion for Computational Linguistics (Volume 1: Long</i>	717
666	Hao Li, Yiqun Zhang, Zhaoyan Guo, Chenxu Wang,	<i>Papers)</i> , pages 15629–15644.	718
667	Shengji Tang, Qiaosheng Zhang, Yang Chen, Biqing	Giovanni Strona, Domenico Nappo, Francesco Boc-	719
668	Qi, Peng Ye, Lei Bai, Zhen Wang, and Shuyue Hu.	cacci, Simone Fattorini, and Jesus San-Miguel-Ayanz.	720
669	2026b. LLMRouterBench: A massive benchmark	2014. A fast and unbiased procedure to randomize	721
670	and unified framework for LLM routing . In <i>Find-</i>	ecological binary matrices with fixed row and column	722
671	<i>ings of the Association for Computational Linguis-</i>	totals . <i>Nature Communications</i> , 5:4114.	723
672	<i>tics: ACL 2026</i> .	Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan	724
673	Tao Li, Jinlong Hu, Yang Wang, Junfeng Liu, and Xue-	Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhou-	725
674	jun Liu. 2025. WebRouter: Query-specific Router	jun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu,	726
675	via Variational Information Bottleneck for Cost-	Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming	727
676	sensitive Web Agent . <i>Preprint</i> , arXiv:2510.11221.	Xiong, Victor Zhong, and Tao Yu. 2024. OSWorld:	728
677	Xunzhuo Liu, Bowei He, Xue Liu, Andy Luo, Haichen	Benchmarking multimodal agents for open-ended	729
678	Zhang, and Huamin Chen. 2026. Adaptive vision-	tasks in real computer environments . In <i>Advances in</i>	730
679	language model routing for computer use agents .	<i>Neural Information Processing Systems 37 (NeurIPS</i>	731
680	<i>Preprint</i> , arXiv:2603.12823.	<i>2024) Datasets and Benchmarks Track</i> .	732
681	Qingyu Lu, Liang Ding, Siyi Cao, Xuebo Liu, Kanjian	Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chun-	733
682	Zhang, Jinxia Zhang, and Dacheng Tao. 2025. Run-	yuan Li, and Jianfeng Gao. 2023. Set-of-Mark	734
683	away is Ashamed, But Helpful: On the Early-Exit	Prompting Unleashes Extraordinary Visual Ground-	735
684	Behavior of Large Language Model-based Agents	ing in GPT-4V . <i>Preprint</i> , arXiv:2310.11441.	736
685	in Embodied Environments . In <i>Findings of the As-</i>	Ke Yang, Yao Liu, Sapana Chaudhary, Rasool Fakoar,	737
686	<i>sociation for Computational Linguistics: EMNLP</i>	Pratik A. Chaudhari, George Karypis, and Huzefa	738
687	<i>2025</i> .	Rangwala. 2025. AgentOccam: A Simple Yet Strong	739
688	Yasmin Moslem and John D. Kelleher. 2026. Dynamic	Baseline for LLM-Based Web Agents . In <i>Interna-</i>	740
689	Model Routing and Cascading for Efficient LLM In-	<i>tional Conference on Learning Representations</i> .	741
690	ference: A Survey . <i>Transactions on Machine Learn-</i>	Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik	742
691	<i>ing Research</i> . Survey Certification.	Narasimhan. 2025. τ-bench: A benchmark for tool-	743
692	Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chi-	agent-user interaction in real-world domains . In <i>In-</i>	744
693	ang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed	<i>ternational Conference on Learning Representations</i> .	745
694	Kadous, and Ion Stoica. 2025. RouteLLM: Learning	Jiayi Yuan, Hao Li, Xinheng Ding, Wenya Xie, Yu-	746
695	to Route LLMs from Preference Data . In <i>Interna-</i>	Jhe Li, Wentian Zhao, Kun Wan, Jing Shi, Xia Hu,	747
696	<i>tional Conference on Learning Representations</i> .	and Zirui Liu. 2025. Understanding and Mitigating	748
		Numerical Sources of Nondeterminism in LLM In-	749
		ference . <i>Preprint</i> , arXiv:2506.09501.	750
		Boyuan Zheng, Boyu Gou, Jihyung Kil, Huan Sun, and	751
		Yu Su. 2024. GPT-4V(ision) is a Generalist Web	752

[Agent, if Grounded](#). In *Proceedings of the 41st International Conference on Machine Learning*.

Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. 2024. [WebArena: A Realistic Web Environment for Building Autonomous Agents](#). In *International Conference on Learning Representations*.

A Task universes and exclusions

Scored sets are the collected tasks minus protocol exclusions written as rules over task configuration: classifieds 224 of 224; VWA-reddit 203 of 205 (one task whose evaluator never checks the requested action, and one cross-site task answerable from memory); shopping 432 of 435. For every site the collected tasks are the benchmark’s tasks minus those it marks as not applicable (WebArena-reddit: 104 of 106). Every analysis records the task set it was built on and fails if denominators differ.

B Agent and run contract

Every condition uses the same agent loop and settings; only the observation mode and backbone vary. Decoding is greedy (temperature 0, at most 4096 new tokens per step). The browser viewport is 1280×720, and the page is observed 0.5 s after each action. The prompt holds the task instruction, the mode’s system prompt, the last eight steps (thought, action, result) and the current observation; the text observation (accessibility tree or marked-element list) is never truncated. The nominal step budget is 30; three consecutive unparsable outputs end an episode. B1 (Qwen3-VL-4B-Instruct) and B2 (Gemma-3-4B-it) run locally from pinned model revisions; B0 (Qwen3-VL-235B-A22B) and B5 (the mid-price GPT-5.6 tier) are reached through a hosted API under a fixed model identifier. Each run uses seed 42, starts from freshly reset sites, and records the code version, configuration and environment with every episode. The exact prompts and configuration files are in the supplementary material.

C The six observation modes

Four modes carry no page screenshot and cross two factors: the text payload (the accessibility tree, or the flattened list of set-of-marks elements) and the prompt family (DOM or SoM). DOM is tree + DOM prompt; P-prompt is tree + SoM prompt; P-text is marks + DOM prompt; P-SoM is marks +

SoM prompt. SoM adds the annotated screenshot to P-SoM; Vision is the raw screenshot with an empty text payload. A task’s own reference image, when it has one, is given in every mode.

D The intent rule

A task is flagged if its intent matches

```
\b(image|picture|photo|screenshot)\b|
\bcolou?r of\b|
\bhow many\b[^\.]{0,40}\bin(?:the|this)\b
```

(case-insensitive) and the task carries no reference image. It flags 71 of 224 classifieds tasks, 63 of 203 VWA-reddit tasks, 56 of 432 shopping tasks and 5 of 104 WebArena-reddit tasks. It was written while auditing reddit failures; classifieds is out-of-sample.

E Harness defects and how cells were filtered

- **GPT-5.6 Vision:** the run used a broken coordinate contract and is excluded.
- **GPT-5.6 step budget:** when the model emitted several valid actions, the parser executed one but booked the step as a free wait. Summed over a whole condition GPT-5.6 had 50–90 such steps (1.2–2.4% of its steps; most of them executed an action, and some episodes reached 31–32 counted steps), B2 had 2–24, and B0 and B1 none. It affects SoM and P-SoM alike.
- **Reddit state:** subscriptions persist across the episodes of a run. Of the reddit successes scored by the subscribed-forum list, six were credited although the episode never visited the target forum, three of them for B2 with DOM; §3 reports the B2 comparison with and without them. The audit cannot detect an episode that reaches an already-subscribed forum and leaves without acting (one such case is confirmed on WebArena-reddit).
- **Shopping search box:** typing into a search box that already holds a query submitted old + new text (“chair” + “chair with wheels” → “chairchair with wheels”) in 43% of such submissions, far more often in text modes than in Vision. A detector reproducing the defect’s per-mode counts marks affected episodes; the *clean* shopping cells drop every task affected in any mode (B0 keeps 368 of 432 tasks, B1 192).

cost	cells	curve max (p)		deployable	
		per-arm	triage	per-arm	triage
billed	all 8	+0.86 (.072)	+1.69 (.001)	+0.23	+0.59
	no B2 6	+1.15 (.071)	+1.85 (.001)	+0.50	+0.39
wall-clock	all 8	+1.28 (.077)	+0.68 (.418)	+0.10	−0.48
	no B2 6	+1.70 (.076)	+0.60 (.619)	+0.18	−0.95
GPU time	all 5	+1.61 (.076)	+1.24 (.387)	+1.47	+0.05
	no B2 3	+2.68 (.076)	+2.06 (.376)	+2.58	−0.56

Table 2: Three-mode routing pooled over cells under three cost bases, with and without the weakest backbone (B2). GPU time exists only for the locally served B1 and B2. Curve max: best point after seeing test outcomes, with its label-shuffle p ; deployable: operating point chosen on training folds. Sources: routing_three_arm and its _wallclock, _gpu_time siblings.

- **Shopping state:** the category grid order differs between runs (42 order-sensitive tasks dropped), and the wishlist is not reset between tasks (4 tasks dropped).

F Six-mode results

Over all six modes, the no-interaction null of §5 is rejected in all three fully rerun cells ($p = 0.0005$, 0.017, 0.0005; Holm), with single-run reliability 0.29, 0.14 and 0.46 and three, six and two runs per task needed to reach 0.5. Across the 24 registered rerun pairs, a rerun changes between 0 and 14% of task outcomes, depending on arm and cell.

G Template and rerun robustness

Averaged over cells, the deployable excess over the frontier with folds grouped by task template is −1.60 (per-arm heads) and −0.84 (triage) points over six modes, and −0.64 and +0.29 over three. Fitting on one run and scoring on its rerun, averaged over the three fully rerun cells, gives −0.90 and −1.60 over six modes and −0.60 and −0.56 over three. Over six modes, every per-cell template-bootstrap interval (90%) includes zero; the three-mode readings are point estimates.

H Per-cell results

Table 3 lists the success rate and cost behind Figures 1 and 4. In shopping the fused mode is cheaper than text: its marked-element list is shorter than the accessibility tree of a product-dense page.

I Estimators

Frontier. For a cell with arms m of mean cost c_m and success rate s_m , $F(c)$ is the upper concave envelope of $\{(c_m, s_m)\}$, cut at the highest-success

cell	n	DOM		SoM		Vision	
		SR	cost	SR	cost	SR	cost
cls B0	224	17.4	6.96	27.2	7.24	25.0	6.48
cls B1	224	6.2	5.95	14.3	6.03	12.5	4.32
cls B2	224	1.3	7.68	2.2	9.08	2.2	7.07
VWA-red B0	203	14.3	10.15	14.8	11.05	7.4	9.81
VWA-red B1	203	5.9	7.33	7.4	8.00	2.5	5.24
VWA-red B2	203	3.9	9.48	1.0	11.16	2.0	6.83
WA-red B0	104	26.9	7.53	22.1	9.11	19.2	8.64
WA-red B1	104	16.3	6.58	13.5	7.94	9.6	4.47
cls B5	224	23.7	13.74	37.1	16.71	—	—
shop B0 clean	368	12.8	12.25	16.8	9.87	16.6	7.75
shop B0 all	432	11.3	11.99	14.8	9.77	14.4	7.48
shop B1 clean	192	5.7	10.60	9.4	7.75	6.8	4.69
shop B1 all	432	4.9	11.11	7.6	8.27	5.6	4.94

Table 3: Success rate (%) and mean billed cost per task (US cents) of the three deployment arms. B0 and GPT-5.6 (B5) costs are API invoices; B1 and B2 are token-priced estimates for local serving, comparable within a cell only. Shopping: clean = tasks on which the search-box defect fired in no mode (Appendix E), all = every scored task; GPT-5.6 Vision is excluded. Sources: routing_three_arm, routing_extension_cells.

arm (constant beyond it) and equal to the cheapest arm’s success below the cheapest cost. Every point of F is reached by sending a random share of tasks to each of two arms. A policy with mean cost c and success s has excess $s - F(c)$.

Router curves. Features are standardised within each training fold and fed to an L2-regularised logistic regression ($C = 1$); predictions are out-of-fold over five folds of distinct tasks. *Per-arm heads:* for each threshold $\tau \in \{0, 0.05, \dots, 1\}$ a task goes to the cheapest arm (by training-fold mean cost) whose predicted success is at least τ , and to the training fold’s best arm if none is. *Triage:* one model predicts whether any arm solves the task; for each quantile $q \in \{0, 0.05, \dots, 1\}$ of its score, tasks below the quantile go to the cheapest arm and the rest to the best arm.

Pooled gain. Within a cell, cost is normalised as $u = (c - c_{\min}) / (c_{\max} - c_{\min})$ over the fixed arms, which makes cells with different cost units comparable without pooling their costs. The gain at u is the envelope of the fixed arms together with the curve’s points minus the envelope of the fixed arms alone, on 101 grid points; the pooled curve is the equal-weight mean over cells and its statistic is the maximum over u . The null permutes each task’s outcomes and costs against its features and refits the whole curve ($B = 1000$; $p = (k + 1) / (B + 1)$); for the pooled statistic each null draw is averaged

over cells before the maximum).

Deployable policy. In each outer fold the threshold (or quantile) is the one with the largest excess over the *training* tasks' own frontier, using in-sample predictions on those tasks; it is then applied to the held-out fold. The five folds' decisions form one policy over all tasks, scored against the full-sample frontier. Intervals resample tasks and rerun the whole pipeline ($B = 1000$, 5th and 95th percentiles); pooled intervals average the cells draw by draw.

No-interaction null and decision study. For the task $\times$ mode $\times$ run array with $r = 2$ runs, the two-way random-effects ANOVA gives $\hat{\sigma}_{\text{err}}^2 = \text{MS}_{\text{err}}$ and $\hat{\sigma}_{\text{int}}^2 = (\text{MS}_{\text{int}} - \text{MS}_{\text{err}})/r$. The null matrices are drawn with curveball trades over the task-by-(mode, run) 0/1 matrix, which preserve every row and column total (four chains, each with 50 trades per row of burn-in and 5 per row between kept draws, 2000 draws in all; for the six-mode analysis a second sampler with four-fold thinning checks mixing). The reliability of a label averaged over k runs is $\hat{\sigma}_{\text{int}}^2 / (\hat{\sigma}_{\text{int}}^2 + \hat{\sigma}_{\text{err}}^2/k)$, and the runs needed for 0.5 are $\lceil \hat{\sigma}_{\text{err}}^2 / \hat{\sigma}_{\text{int}}^2 \rceil$.